

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

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Author contribution statement

G.S. designed the dataset and the label scheme, implemented the processing pipeline and the quality-control procedures, and drafted the manuscript. A.S., A.T., M.K., R.C., D.H., F.M., H.S., M.S., S.S., M.M. and J.K. performed annotation correction and quality-control review of the released segmentations under supervision. M.I. and N.A. performed the radiological reads — each graded every Castellvi case independently and the two resolved their disagreements by consensus — and the adjudication of candidate instrumentation. [BEFORE SUBMISSION: confirm each contribution individually with the author concerned and expand to CRediT roles.] All authors read and approved the final version of the manuscript.

Conflict of interest statement

The authors have no relevant conflicts of interest to disclose.

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Data and code availability (unblinded)

Dataset, archived before submission: <https://doi.org/10.5281/zenodo.22139642> (concept identifier, resolves to the current version); the version described in the manuscript is v8, <https://doi.org/10.5281/zenodo.22293878>.

Code: <https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K>, tagged at the release commit.

Mirror: <https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K> — a convenience mirror, not the archive of record.

Ethics

This work uses publicly available, de-identified imaging and annotations derived from it. The Wayne State University Institutional Review Board issued a Notice of IRB Administrative Determination of Non-Human Participant Research on 2 September 2026 (WSU IRB HPR Number 2026-269, “Open Spine Consortium: Secondary Analysis of Publicly Available, De-Identified Imaging Datasets and the Derived Annotations, Software, and Models Produced From Them”), finding that the project does not constitute human participant research under the Common Rule at 45 CFR 46 and that IRB review and oversight are not required.

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

[Author names removed for double-anonymized review]

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Purpose: Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present. Counting down from C2, the reference standard for numeration¹, is impossible in an abdominal field of view, which leaves a specific ambiguity: four rib-free lumbar vertebrae may mean an L1 bearing a lumbar rib or an L5 assimilated to the sacrum, and six may mean a true sixth lumbar vertebra or a T12 whose ribs are aplastic². CTSpinoPelvic1K is built to test whether local morphology can resolve them without counting from the C2 anchor. It provides 802 CT records anchoring each lumbar level on the lowest rib-bearing vertebra above and the first sacral segment below, with radiologist-sourced spine *and* pelvic annotation on one frame, per-level ribs and femora. Classes for a sixth lumbar vertebra, a thirteenth thoracic vertebra, a separate first sacral segment and lumbar ribs cover the anomalies.

Acquisition and Validation Methods: Records pair TCIA CT COLONOGRAPHY imaging with CTSpine1K³ vertebral and CTPelvic1K⁴ pelvic labels on the acquisition with highest bone coverage, under a VerSe-native scheme. Validation: geometric invariants (802/802 pass); rib-vertebra incidence across 5,749 evaluable ribs (2 offsets, 0.035%); and agreement of spinopelvic measures with published values.

Data Format and Usage Notes: Gzipped NIfTI image/label pairs sharing one affine, canonicalized to PIR, with frozen patient-grouped LSTV-stratified five-fold splits and a reference loader; archived under a persistent identifier, [DOI redacted for review].

Potential Applications: Testing whether a network can classify a vertebra from local features alone; updating textbook morphometry drawn from small cadaveric series; spinopelvic assessment at the lumbosacral interface; variant studies; opportunistic screening. *Limitations:* thoracic ground truth is field-of-view limited; postural angles supine; no held-out test set; the rib layer is a triaged-review pseudolabel; Castellvi grades are a two-reader consensus.

6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the two
9 ends of the lumbar column — the thoracolumbar transitional vertebra (TLTV) above, the
10 lumbosacral transitional vertebra (LSTV) below — and they disturb identification in two
11 ways at once. They change what the border vertebra and its ribs *look like*: a thoracolumbar
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral
13 one may be partly or completely assimilated to the sacrum, with an anteriorly wedged body,
14 a reduced disc beneath it and hypoplastic facet joints, or conversely separated from it with
15 a squared body, lumbar-type facets and a full-height disc². They also change *how many*
16 vertebrae each region contains — eleven or thirteen thoracic, four or six lumbar.

17 Either change alone would be tractable. Together they make it difficult to state defini-
18 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum may
19 mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true sixth
20 lumbar vertebra or a T12 whose ribs are aplastic. Each pair yields the same count, and the
21 morphology that would separate the two readings is not decisive to a human reader in every
22 case. Whether it nonetheless differs enough for a learned classifier to separate them, or at
23 least to assign each reading a probability, is the question this release is built to support.
24 Absent that, what remains is the sequence — conventionally established by counting down-
25 ward from C2, the reference standard for numeration¹, and unavailable in any spine-limited
26 acquisition. LSTV is common, reported between 4% and 30% depending on definition, and
27 wrong-level surgery is its most serious consequence.

28 Existing public collections cannot address this, and size is not the reason (Table I). Spine
29 datasets omit the pelvis; pelvic datasets do not number the vertebrae; and TotalSegmentator,
30 the whole-body scheme, has no class for a sixth lumbar vertebra or for a rib on a lumbar
31 vertebra. A six-lumbar spine therefore cannot be recorded in any of them, however good
32 the segmentation. VerSe^{5,6} makes the point most clearly (Table I): it deliberately enriched
33 for anatomical variants and graded them by Castellvi, yet left the graded vertebrae fused to
34 the sacrum unsegmented.

35 CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and
36 gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra,

37 and a rib borne by a lumbar vertebra. A scheme without those classes must record such
38 anatomy as something it is not.

39 *Where the material is.* Everything described here is v8 of a single archived dataset, de-
40 posited at [DOI redacted for review]. Citations should use [DOI redacted for review], the per-
41 manent identifier for the dataset across all of its versions, which resolves to whichever version
42 is current; the version identifier above pins the exact release measured in this manuscript.
43 Documentation, the label scheme, the reference loader and the per-record quality-control
44 tables are distributed with the archive and are described in Sec. [III](#).

45 **I.A. Vertebra labeling when the scan cannot settle the count**

46 *The limitation, stated plainly.* No scan in this corpus contains C2 or the whole cervical
47 spine, so the conventional count cannot be performed. That is a property of abdominal
48 imaging rather than of the annotation, and it means a variant at the thoracolumbar junction
49 cannot be *resolved by counting* here at all: a thirteenth thoracic vertebra, a lumbar rib, and
50 a *stump rib* — a hypoplastic twelfth rib, one of the cardinal indicators of a thoracolumbar
51 transitional vertebra — produce overlapping appearances.

52 *What the release does instead.* No vertebral label is asserted where it cannot be supported.
53 Every quantity is expressed against the two landmarks present in essentially every abdominal
54 scan — the sacrum below and the lowest rib-bearing vertebra above — so the description is
55 positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from
56 the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process
57 by its span and its distance from the ala. None of these changes according to whether a
58 reader would have said L5 or S1, so cases that disagree about the name remain comparable.
59 Where the junction is in view the count is determined and the conventional name follows,
60 and the two descriptions agree. The positional description is, by construction, blind to the
61 ambiguity: an L1 bearing a lumbar rib and an L5 assimilated to the sacrum both leave
62 four rib-free vertebrae above the sacrum, and a true sixth lumbar vertebra and a T12 with
63 aplastic ribs both leave six. The count alone cannot separate the readings.

64 *But the vertebrae themselves are known to differ.* Counting is not the only route to an
65 identity. Level-specific morphometry is long established: pedicle width and height change
66 systematically from the thoracic to the lumbar spine,⁷ vertebral body, endplate and canal

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it⁶.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K ³	1,005									
CTPelvic1K ⁴	1,184									
VerSe ⁵	374									
RibSeg v2 ¹²	660									
TotalSegmentator ¹⁶	1,204									
CTSpinoPelvic1K	802									

67 dimensions differ by level,^{8–10} and the transverse process and the iliolumbar ligament mark
 68 the last lumbar vertebra independently of any count.^{2,11} If those differences hold at the
 69 boundary itself, the phenotype is decidable from the vertebra in front of the reader without
 70 any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with
 71 each patient’s own size divided out and cross-validation grouped by case, gives an area under
 72 the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is deliberately removed,
 73 because a classifier given raw millimeters separates thoracic from lumbar immediately and
 74 has learned nothing that helps at the junction, where the two are nearly the same size.

75 That is reported as a property of the released measurements rather than as a method:
 76 the information needed to settle these variants is present locally.

77 II. ACQUISITION AND VALIDATION METHODS

78 II.A. Sources, and why a crosswalk was necessary

79 Both CTSpine1K³ and CTPelvic1K⁴ draw part of their imaging from the TCIA CT
 80 colonography collection^{13,14}, and the vertebral and pelvic annotations in both were pro-
 81 duced under board-certified radiologist supervision. Neither released the mapping from an
 82 annotation to the specific CT series it was drawn on.

83 That mapping matters because every patient in CT COLONOGRAPHY was scanned
 84 *twice*, prone and supine, and an annotation carries a patient identifier and nothing finer.
 85 Turning a patient from supine to prone alters lumbar lordosis and the alignment of each

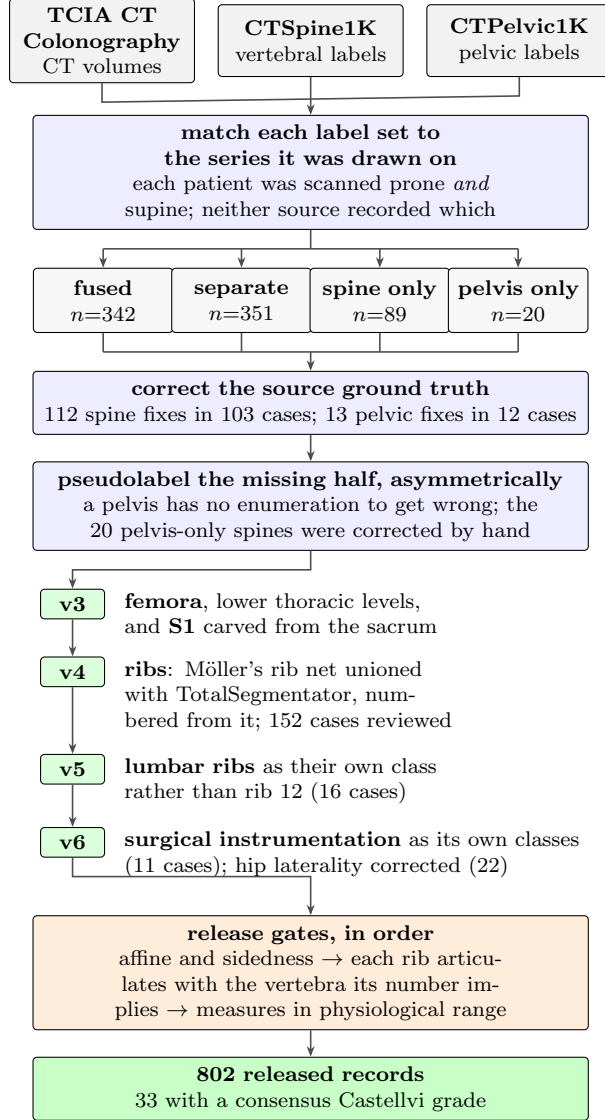


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

86 segment against the next, so outlines drawn on one series are the wrong *shape* for the other
 87 and no rigid realignment recovers them. Such a mask looks competent in isolation and
 88 reveals itself only overlaid on the image. The crosswalk (Fig. 1) therefore resolves each
 89 annotation to a patient and then, within that patient's volumes, to the series whose bone
 90 *agrees* with it: candidates are thresholded at bone attenuation and scored by overlap with
 91 the mask.

92 II.B. Fused and split records

93 CTSpine1K and CTPelvic1K each annotated a patient independently, so their labels are
94 not guaranteed to come from the same CT series (Fig. 1). This turned out to matter for
95 351 patients, nearly half the cohort: the two collections had labeled different acquisitions of
96 the same person, one prone and one supine, with neither recording which. Because posture
97 changes spinal alignment, a spine label from one acquisition cannot simply be paired with a
98 pelvic label from the other. These 351 records are released but held out of any analysis that
99 assumes a single posture. Two labeled acquisitions of the same patient in different postures
100 are, instead, a resource for studying postural alignment, not a defect.

101 II.C. Densification, and why it is asymmetric

102 A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing
103 structures were pseudolabeled, asymmetrically (Fig. 1). Pelvis onto spine-only records is the
104 safe direction: the sacrum and innominate bones carry no enumeration to get wrong, and
105 quality is reported as held-out performance on the *pelvic_only* records, withheld from training
106 for that purpose. The reverse direction carries the whole problem this dataset addresses,
107 since a pseudolabeled spine must commit to a count and on a transitional vertebra commits
108 to the commoner reading; those 20 records were corrected by hand, tractable at that scale.

109 II.D. Ribs

110 No public rib annotation covers this cohort, and the two available automatic sources fail
111 in complementary ways.

112 Möller’s binary rib network¹⁵ reports high accuracy, and its authors show that stump ribs
113 can be classified from a partial field of view; that result concerns the morphological features
114 computed on a rib once segmented. On this corpus the segmentation itself failed on ribs
115 that were *partially outside* the field of view: a rib crossing the boundary of an abdominal
116 acquisition was liable to be missed altogether rather than segmented to the edge. This is
117 not a limitation reported by its authors: their evaluation excluded subjects in which the last
118 rib was not visible, and their failure analysis attributes errors to costal-process fusion, so it
119 is recorded here as an observation from this cohort. Since this is a colonography cohort, the

120 lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

121 TotalSegmentator¹⁶ does segment clipped ribs and supplies the per-rib numbering that a
122 binary network cannot. Its characteristic failure is at the other end of the bone: the most
123 medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral
124 joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra
125 it articulates with, so the rib–vertebra incidence check (the quality-control gate on the rib
126 class labels, which matches each rib to the vertebra its number implies) has nothing to test.

127 The two were therefore unioned (Fig. 1): TotalSegmentator supplies numbering and the
128 medial-clipped cases, Möller supplies detection where its output is more complete, and the
129 union carries TotalSegmentator’s per-level identities.

130 The result is a pseudolabel whose review was triaged rather than exhaustive, which
131 is worth stating plainly because a layer described only as “human-corrected” invites the
132 reader to assume every bone was looked at. A label-based rule flagged every record in
133 which one rib bone carried more than one identifier, or in which a rib failed to reach the
134 vertebra it articulates with, since a single bone carrying two numbers is a numbering error by
135 construction, whatever the anatomy. The rule selected 152 records; 148 were reviewed and
136 corrected, and 4 were not and are identified in the release. Reviewers were medical students
137 working through a student research consortium¹⁷, trained against a written labeling protocol,
138 with every correction recorded against the annotator who made it. Records passing the rule
139 were not individually inspected, so the layer only guarantees freedom from the failure modes
140 the rule detects, no more. The residual rib–vertebra offsets reported in Sec. II.G are the
141 independent check on what the rule missed.

142 II.E. Label scheme and counting anchors

143 The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12
144 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier
145 above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs
146 (34–45 left, 46–57 right), lumbar ribs (74–75) and surgical hardware (76–82). Identifiers
147 58–73 are unassigned: a soft-tissue block once reserved there was never populated and is
148 retired.

149 Two anchors make the count decidable, and Fig. 2 shows both: the lowest rib-bearing

150 vertebra above, **S1** carved from the sacrum below. Each earns its place by what it supports
151 rather than by being an endpoint — S1 supplies the sacral endplate landmark that sacral
152 slope and pelvic incidence are measured from, and with both brackets present L5 and L6
153 are distinguishable where a spine-limited view alone would not be.

154 *How S1 is cut.* The sacrum’s outer boundary is the source annotation and is never altered;
155 S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the
156 sacrum’s own cranio-caudal axis, made orthogonal to a left–right axis measured directly
157 rather than assumed, so as to remove the patient’s rotation within the scanner. That
158 rotation is not negligible: median 4.5°, maximum 16.1°, and 508 of 801 records above 3°.
159 The level of the cut preserves the sub-division volume of the preceding release, so the plane
160 changes the shape and orientation of the S1/S2 boundary and not how much of the sacrum
161 is called S1. Per-record geometry ships with the archive.

162 Neither anchor is novel on its own: TotalSegmentator¹⁶ carries a separate S1 class and
163 per-level ribs, and VerSe⁵ carries L6. Because this release uses the VerSe identifiers, its L6
164 *is* VerSe’s, interoperable rather than local to this work. L6 is retained not for novelty but
165 because a scheme without it cannot record a six-lumbar spine at all, and must renumber the
166 column or drop a level to fit.

167 **A rib on a lumbar body receives its own class**, and this is the one class here with no
168 published counterpart. A lumbar rib and a stump rib are the same object under two counts,
169 so assigning it a number would commit the label to one reading of an enumeration anomaly.
170 A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must
171 either call it rib 12 — asserting the vertebra beneath it is thoracic, the very question at
172 issue — or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and
173 VerSe’s T13 is a vertebra rather than a rib. 16 released records carry a lumbar rib and 18
174 carry an L6.

175 Figure 2 shows why the interval is the primitive and the label is derived. Among the
176 records with four rib-free vertebrae, some reach that count through a lumbar rib and some
177 without one; the two are indistinguishable by the count alone and are distinguished here
178 only because rib presence is itself labeled.

179 Sixteen records carry a lumbar rib in the released volumes: thirteen bilateral and three
180 unilateral, all three on the right. That laterality is reported because it is what the release
181 contains and not as a finding — sixteen records cannot support a statement about side

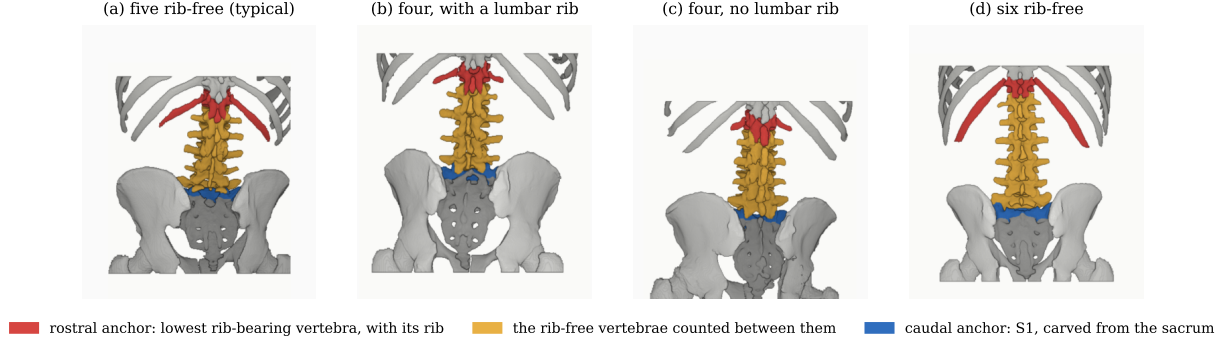


FIG. 2 The two anchors, from the released labels. Red, lowest rib-bearing vertebra and its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four.

182 preference, and none is made.

183 Eighteen records carry an L6. Fourteen are labeled lumbarization, two sacralization,
 184 one semi-sacralization and one normal, which is worth stating because it shows the two
 185 annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and
 186 still be read as a sacralization, or as unremarkable, depending on where the reader started
 187 the count.

188 Both counts are derived from the released label volumes, and so are the manifest fields
 189 that report them. That is a correction rather than a design note: through v5 the `has_l6`
 190 flag was true for exactly one record, which contains no L6, and false for all eighteen that
 191 do, while `has_lumbar_rib` was false everywhere and `n_lumbar_labels` read zero in 799 of
 192 802 records. A reader selecting the six-lumbar cases on those fields would have received one
 193 record that is not one, and a reader selecting lumbar ribs an empty set, with nothing to
 194 indicate either. The fields are rebuilt for this release by counting identifiers 25 and 74–75 in
 195 each volume, and `lumbar_rib_side` is added alongside them.

196 II.F. Computational tools

197 *Access.* Every processing step described above is performed by code released under an
 198 open-source licence and archived with the dataset; no step depends on software that is
 199 not publicly obtainable. The release archive carries the reference loader, the label-scheme
 200 definition and the quality-control scripts that generated the tables in this manuscript.

201 *Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-

level rib numbering used TotalSegmentator¹⁶ (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller’s¹⁵ released weights, applied through the nnU-Net v2 framework¹⁸. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold that never trained on it, so no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (<https://huggingface.co/anonymous-mlhc/spinopelvic-seg-checkpoints>). Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image’s grid and affine exactly.

Generative artificial intelligence. Assistance from large language models was used for manuscript preparation — drafting and editing of text, and generation of analysis and figure code — under author review. It was not used to generate, annotate or interpret any imaging data, and no scientific claim in this manuscript originates from it. All numbers reported here are computed by the released code from the released volumes.

II.G. Validation

Validation is layered, and the ordering matters: measurements computed on a corpus that has not established internal consistency do not look broken, they look finished.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. The check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed a label away from its image, a failure invisible downstream.

The check compares *each sided pair separately*, and both simpler alternatives miss cases it catches. Testing the ribs alone passed all 802 records while four of them carried `left hip` on the patient’s right. Pooling all sided structures into one test recovered three of those four and missed the fourth, because hips and femora outweigh the ribs by voxel count, so a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against

the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered. The denominator is stated because quoting the two residual offsets against all 11,548 ribs would halve the rate by counting ribs the check never examined — and because more than half the ribs in a screening abdominal CT being unevaluable is itself a fact about the corpus worth reporting. Both offsets are field-of-view truncations on individually characterized cases, one at +1 level and one at −1, and are reported rather than suppressed: a threshold tuned until the count reaches zero yields a cleaner number and a less informative one.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra — not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 74 or 75 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*¹⁹ (Table II). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

II.H. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade²⁰ in the released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier. This is released as an annotation layer because the grade and the count are *different axes* and the dataset is built to show that they are: grade IIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A corpus that recorded only the count would describe those seven as unremarkable.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the classification turns on — bony fusion against articulation without it² — is the one carrying clinical weight, and it is where the disagreements lay: four of the six were

TABLE II Derived measures against a standing reference ($n = 260$)¹⁹; this cohort is supine.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centered near 0

III against II.

II.I. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**: a cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does. An unrecorded instrumented case is therefore a false positive on the central measure.

II.I.0.a. Detection over-calls by a factor of five. A 2500 HU threshold inside a shell around the labeled skeleton flags 52 of the 802 records. One of the two radiology resident physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Attenuation does not separate the two groups (both saturate); site and volume do, every confirmed implant at 2,586 mm³ or above and every rejection at 1,768 mm³ or below. Instrumentation prevalence is 1.4%, not the 6.5% the threshold alone reports.

II.I.0.b. The subtype block had to be extended. Identifiers 76–79 name spinal instrumentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis** were added: osteosynthesis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule calls a femoral stem a rod. The confirmed cases are 8 arthroplasties, one osteosynthesis, one sacroiliac fixation and one pair of interbody cages (Fig. 3).

II.I.0.c. Metal outranks bone, and that is a subtraction. In every confirmed case the implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than

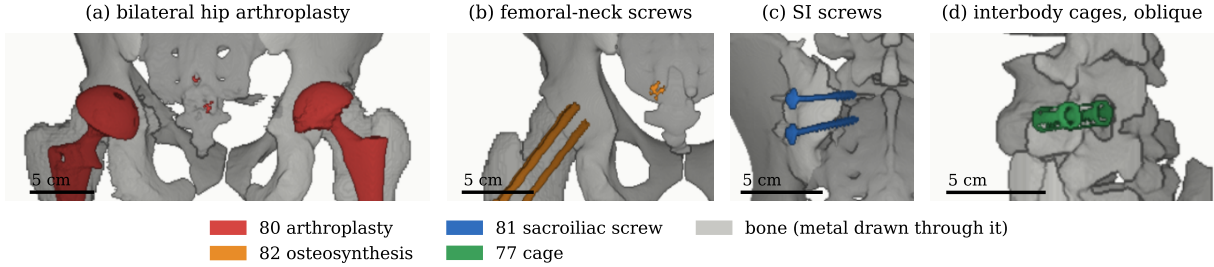


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 80, $n = 8$. (b) Femoral-neck screws, 82, $n = 1$. (c) Sacroiliac screws, 81, $n = 1$. (d) Interbody cages, 77, $n = 1$. Bar, 5 cm.

284 adding them. Pelvic incidence and tilt are measured from the femoral head center, and in
 285 8 of these cases that head is an implant: any spinopelvic parameter previously computed
 286 from them was measured on metal.

287 III. DATA FORMAT AND USAGE NOTES

288 Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canon-
 289 icalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy
 290 prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the
 291 downstream tooling expects.

292 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the
 293 data as `splits_5fold.json`. The five validation folds partition all 802 records, so the
 294 release provides cross-validation and *not* a held-out test set. Users reporting a single held-
 295 out number should carve one out and state which records it holds, because the folds shipped
 296 here will not do that job.

297 Stratification is on the transitional subtype rather than on a binary LSTV flag, and it
 298 enforces a minimum of each rare subtype in every validation fold: each fold’s validation
 299 set carries three sacralization and three lumbarization records, so no fold is evaluated on a
 300 cohort missing the class the dataset exists for. With 15 lumbarization and 15 sacralization
 301 records across 802, that is the practical limit of what stratification can guarantee, and it is
 302 why a fold-level metric on the rare classes carries an error bar far wider than its own decimal
 303 places.

TABLE III Data types and metadata, and the records populating each, from the released manifest.

Field	Records	Available
Identifier, image path, label path, configuration, match type	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class	802	100.0%
Spine and pelvic annotation origin	802	100.0%
Instrumentation and partial-annotation flags	802	100.0%
Image/label alignment check	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	33	4.1%

Table III lists the key data types and metadata against the fraction of records for which each is populated.

The archive of record is Zenodo ([DOI redacted for review]). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value, a recorded value and not a missing one, and 53 carry no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from sex-stratified measures (a limitation of those measures rather than of the cohort), stated because folding eleven people into “unrecorded” would misdescribe the source. The lower age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer to a reference range than a surgical series would.

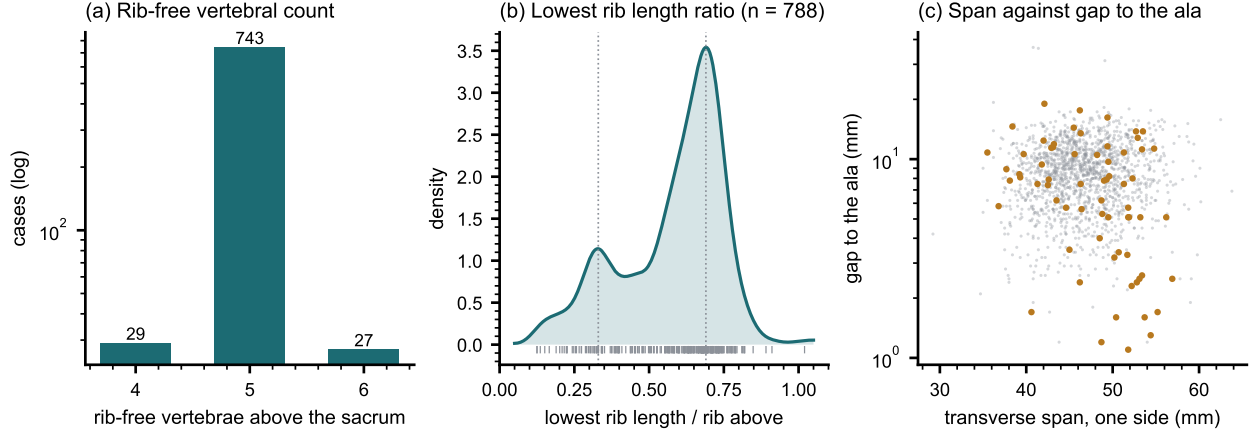


FIG. 4 Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b) Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to the ala; transitional labels marked.

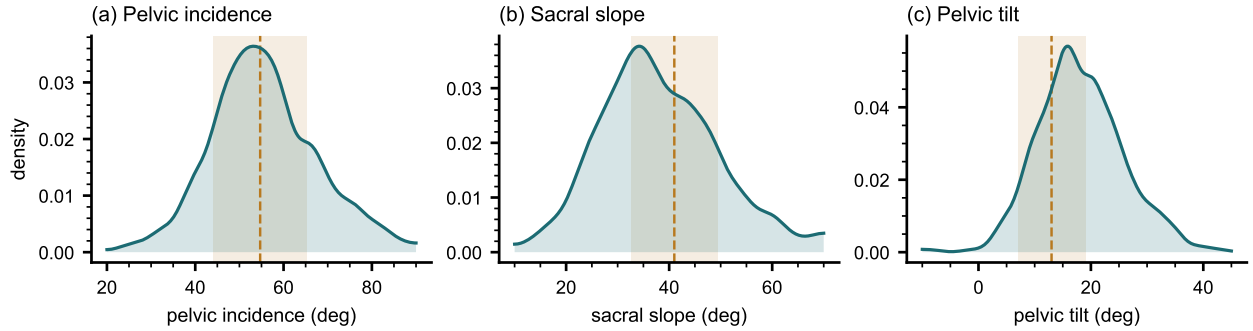


FIG. 5 Spinopelvic measures against standing reference bands (mean $\pm$ SD)¹⁹.

Count-free measures (Fig. 4). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal.

Level gradients and spinopelvic measures (Fig. 5). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

Opportunistic measures. Every scan carries a vertebral bone density measurement at no additional dose.²¹ L1 trabecular attenuation is reported at the published standard site.

326 V. POTENTIAL APPLICATIONS AND LIMITATIONS

327 The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced
328 annotation on one coordinate frame in 802 records, making sacral morphology measurable
329 against the lumbar column rather than in isolation — the sacral endplate and its slope, the
330 alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry
331 a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and
332 femoral heads rather than from landmarks placed on a radiograph, so it is reproducible from
333 the labels themselves.

334 The transitional layer serves that use rather than competing with it: no trajectory can be
335 planned across a junction whose segments cannot be named. Level-numbering benchmarks,
336 variant studies and opportunistic screening follow from the same annotation.

337 V.A. Three uses the released measurements support

338 *Naming a level from its own geometry.* Wrong-level spine surgery runs at roughly one
339 in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this
340 cohort was assembled around.²² The failure is structural rather than careless: the count
341 is ambiguous exactly where the variant sits. Per-level body height, canal width, endplate
342 width, transverse-process span and wedge ratio are released for T11–L5 across all 802 records,
343 and because the labels were assigned against the twelfth rib rather than by enumeration, a
344 model trained on them is not learning the convention that fails.

345 *Reference morphometry with its spread.* The values a surgeon consults for vertebral
346 body height, canal dimensions and pedicle width come from cadaveric series of a few dozen
347 specimens, plotted as single values with no indication of spread.¹⁰ They cannot tell a reader
348 whether the patient in front of them is unusual, because they never show what usual looks
349 like. The same measures are given here from 802 records with the distribution attached
350 (Fig. 6), reproducing two properties of the classical figures — body height crosses over,
351 dorsal exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens
352 steeply into the lower lumbar spine — and adding the width of each distribution, which is
353 the part needed to judge an individual. A larger n sharpens a distribution without changing
354 the population it came from, and this one is a screening cohort aged 50 and over.

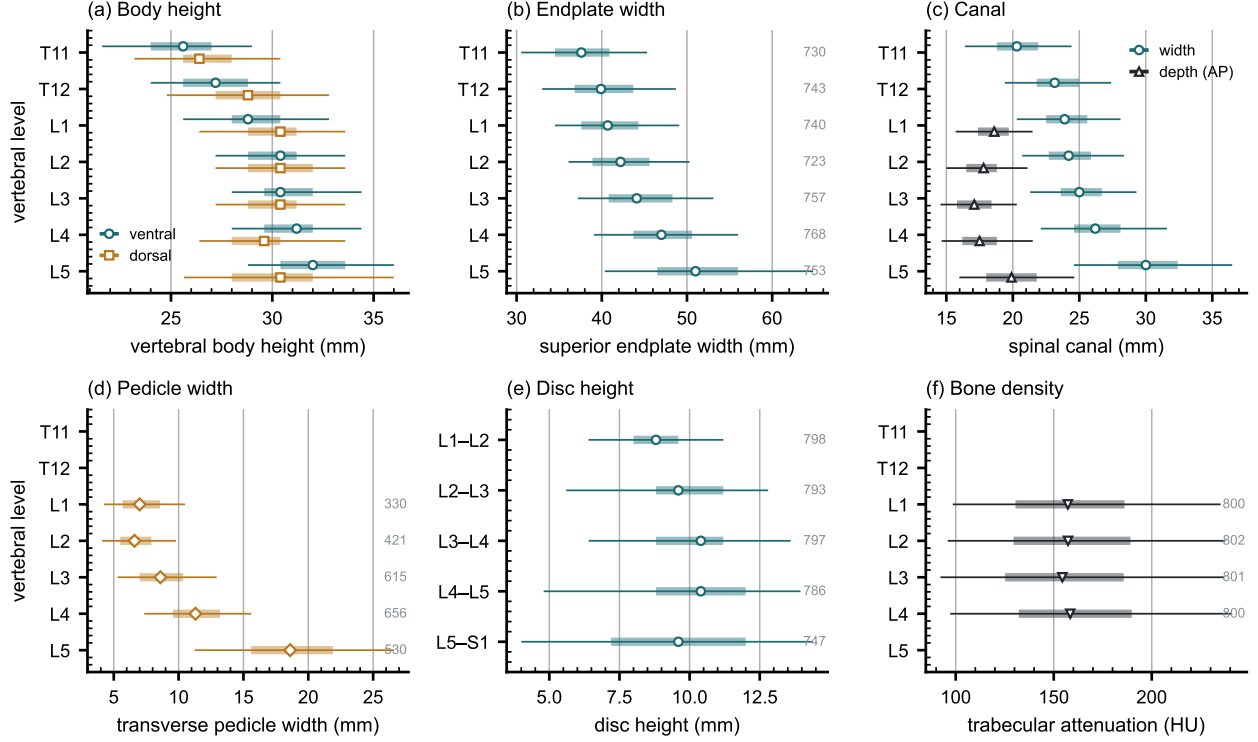


FIG. 6 Morphometry by level: median, interquartile range, 5th–95th percentile, n at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height. (f) Trabecular attenuation.

Patient-specific models. Planning is moving toward patient-specific biomechanical models rather than population norms.²³ Such a model needs spine, pelvis and femoral heads in one frame, which is this release’s construction. In 351 patients the annotation sits on two acquisitions in different positions, giving a within-patient change in alignment against which a predicted postural response can be checked.

Its limitations are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited (Fig. 7) and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

Declared but empty, and confirmed so. The partial-annotation sentinel (255) is declared in the scheme and occurs in no record — established by counting identifiers across all 802 released volumes rather than asserted. It is retained because the protocol is part of the

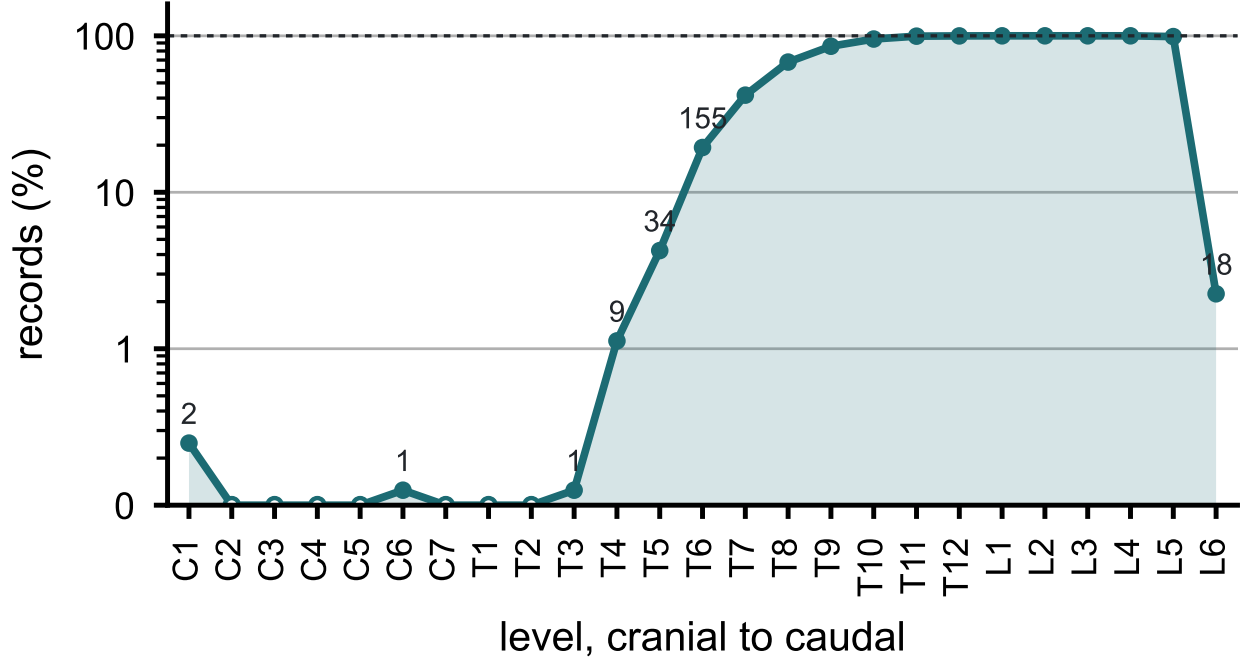


FIG. 7 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

369 scheme and a later release may ship a partially traced record. The hardware identifiers were
 370 in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented
 371 records described in Sec. II.I.

372 *Strength of the labels varies by structure, and the release does not average over that.*
 373 Vertebral labels derive from radiologist-supervised source annotations, corrected where those
 374 sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib
 375 layer is a pseudolabel whose human review was triaged by an automated rule rather than
 376 exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional
 377 labels derive from two independent sources and are not uniformly adjudicated — which is
 378 precisely why the measures reported here are count-free — and the Castellvi grades are
 379 a consensus of two radiology resident physicians. The sacral sub-division inherits its level
 380 from an automatic method, and in 100 records that level places S1 above half the sacrum
 381 or below fifteen percent of it, which no first sacral segment is; those records are flagged in
 382 the released quality-control table and should be filtered before any measurement depending
 383 on the sacral endplate.

384 *Structures are not universally present.* The same census shows the release is not uniformly
 385 complete. Sacrum, hips and femora are present in all 802 records. One record carries no

386 S1, one lacks T12, and nine lack an L5 identifier — in every case because the structure
387 lies outside the field of view or, for L5, because the lowest lumbar segment is labeled L6 or
388 incorporated into the sacrum. Users filtering on the presence of the caudal anchor should
389 filter explicitly rather than assume it.

390 VI. DISCUSSION

391 *Comparison with related material.* Against the collections in Table I, the contribution
392 is the crosswalk placing spine and pelvis on the same series, together with classes for the
393 anatomy an enumeration anomaly actually produces. VerSe⁵ comes closest in intent: it
394 also enriched its cohort for transitional anatomy and graded each case by Castellvi. The
395 difference is what a graded case leaves behind. VerSe names the anomaly but does not
396 segment the sacrum, so the fused or articulating structure the grade describes has no mask
397 to go with it. This release segments that structure directly, as L6 or as a separately carved
398 S1, so a user has the anatomy itself to measure rather than only a grade that describes it.
399 TotalSegmentator¹⁶ invites a different comparison, since it segments far more of the body
400 than any single source used here. But its label scheme simply has no class for a sixth lumbar
401 vertebra or for a rib on a lumbar body. A six-lumbar spine cannot be recorded in it: the
402 column must be renumbered or a level dropped to fit. What this release adds, then, is not
403 more anatomy per scan (TotalSegmentator already covers more), but the specific classes an
404 enumeration anomaly needs in order to be recorded accurately.

405 *Limitations.* The cohort comes from one acquisition protocol of a colorectal-screening
406 population aged 50 and over, so how well it generalizes to a pathologic population is untested.
407 The rib layer’s human review was triaged by an automated rule rather than performed on
408 every bone, so it guarantees only the absence of the failure modes that rule detects.

409 *Future directions.* The version history shows the scheme being extended more than once:
410 femora and the S1 anchor, then per-level ribs, then lumbar ribs and surgical hardware, were
411 each folded into the existing release rather than requiring a new dataset. The same path is
412 open going forward: imaging from beyond this cohort’s single protocol, and the C2 anchor
413 this acquisition cannot supply, are the natural next additions.

414 **VII. CONCLUSION**

415 CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame
416 in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of
417 its own rather than recording it as something it is not. Because both counting anchors are
418 explicit, a level can be identified from a field of view that cannot support the conventional
419 count downward from C2, which is every record here.

420 **DATA AVAILABILITY**

421 The release described here is v8, permanently archived at [DOI redacted for review]; the
422 concept DOI [DOI redacted for review] resolves to the current version. The archive carries
423 the loader, the label scheme, the quality-control tables and the analysis code under an open-
424 source licence. The repository URL identifies the authors; it is on the title page and will be
425 restored here for the camera-ready version.

426 **CONFLICT OF INTEREST**

427 The authors have no relevant conflicts of interest to disclose.

428 **ETHICS**

429 This work uses publicly available, de-identified imaging and annotations derived from it.
430 The authors' institutional review board determined that it is not human participant research
431 under 45 CFR 46 and requires no oversight.

432 **REFERENCES**

433 ¹J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and
434 associated vertebral numeration, *Eur. Spine J.* **27**, 995–1004 (2018), doi:10.1007/s00586-
435 018-5554-8.

²G. P. Konin and D. M. Walz, Lumbosacral transitional vertebrae: classification, imaging findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010), doi:10.3174/ajnr.A2036.

³Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.

⁴P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.

⁵A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021), doi:10.1016/j.media.2021.102166.

⁶H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-021-01060-0.

⁷M. R. Zindrick et al., Analysis of the morphometric characteristics of the thoracic and lumbar pedicles, *Spine (Phila. Pa. 1976)* **12**, 160–166 (1987), doi:10.1097/00007632-198703000-00012.

⁸M. M. Panjabi et al., Thoracic human vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **16**, 888–901 (1991), doi:10.1097/00007632-199108000-00006.

⁹M. M. Panjabi et al., Human lumbar vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **17**, 299–306 (1992), doi:10.1097/00007632-199203000-00010.

¹⁰E. C. Benzel, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015.

¹¹R. J. Hughes and A. Saifuddin, Numbering of lumbosacral transitional vertebrae on MRI: role of the iliolumbar ligaments, *AJR Am. J. Roentgenol.* **187**, W59–W65 (2006), doi:10.2214/AJR.05.0415.

¹²L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024), doi:10.1109/TMI.2023.3313627.

¹³K. Smith et al., Data from CT COLONOGRAPHY, The Cancer Imaging Archive, 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.

- ¹⁴K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.
- ¹⁵H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.
- ¹⁶J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.
- ¹⁷[Reference to the authors’ own work, removed for double-anonymized review.]
- ¹⁸F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat. Methods* **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.
- ¹⁹R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, *J. Bone Joint Surg. Am.* **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.
- ²⁰A. E. Castellvi, L. A. Goldstein, and D. P. K. Chan, Lumbosacral transitional vertebrae and their relationship with lumbar extradural defects, *Spine (Phila. Pa. 1976)* **9**, 493–495 (1984), doi:10.1097/00007632-198407000-00014.
- ²¹P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley, Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications, *Ann. Intern. Med.* **158**, 588–595 (2013), doi:10.7326/0003-4819-158-8-201304160-00003.
- ²²N. E. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery, *Surg. Neurol. Int.* **12**, 286 (2021), doi:10.25259/SNI_402_2021.
- ²³E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, *J. Clin. Med.* **15**, 2584 (2026), doi:10.3390/jcm15072584.

Supplementary Material

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

Table S1: Every identifier populated in the release and the number of the 802 records carrying it, counted from the released label volumes. Identifiers 58–73 are unassigned and occur nowhere.

id	class	records	%	id	class	records	%
1	C1	2	0.2	37	rib_left_4	65	8.1
6	C6	1	0.1	38	rib_left_5	316	39.4
10	T3	1	0.1	39	rib_left_6	635	79.2
11	T4	9	1.1	40	rib_left_7	780	97.3
12	T5	34	4.2	41	rib_left_8	800	99.8
13	T6	155	19.3	42	rib_left_9	802	100.0
14	T7	335	41.8	43	rib_left_10	802	100.0
15	T8	545	68.0	44	rib_left_11	802	100.0
16	T9	688	85.8	45	rib_left_12	787	98.1
17	T10	765	95.4	48	rib_right_3	5	0.6
18	T11	798	99.5	49	rib_right_4	62	7.7
19	T12	801	99.9	50	rib_right_5	310	38.7
20	L1	802	100.0	51	rib_right_6	624	77.8
21	L2	802	100.0	52	rib_right_7	773	96.4
22	L3	802	100.0	53	rib_right_8	800	99.8
23	L4	802	100.0	54	rib_right_9	801	99.9
24	L5	793	98.9	55	rib_right_10	801	99.9
25	L6	18	2.2	56	rib_right_11	802	100.0
26	sacrum	802	100.0	57	rib_right_12	788	98.3
29	S1	801	99.9	74	rib_left_lumbar	13	1.6
30	left_hip	802	100.0	75	rib_right_lumbar	16	2.0
31	right_hip	802	100.0	77	hardware_cage	1	0.1
32	femur_left	802	100.0	80	hardware_arthroplasty	8	1.0
33	femur_right	802	100.0	81	hardware_si_screw	1	0.1
36	rib_left_3	5	0.6	82	hardware_osteosynthesis	1	0.1